

Rajiv Sambharya

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Academic Positions

- **Assistant Professor, Texas A&M University** August 2026-
Industrial and Systems Engineering
- **Postdoctoral Researcher, University of Pennsylvania** Sept 2024-July 2026
Electrical and Systems Engineering
Supervision: George Pappas

Education

- **Ph.D., Princeton University** Sept 2019-Sept 2024
Operations Research and Financial Engineering
Thesis: Learning to Accelerate Optimization Algorithms with Guarantees
Supervision: Bartolomeo Stellato
- **M.Sc. University of California - Berkeley** Sept 2017-Sept 2018
Electrical Engineering and Computer Science
Supervision: Laurent El Ghaoui
- **B.Sc. University of California - Berkeley** Sept 2013-Sept 2017
Electrical Engineering and Computer Science

Research Interests

- **Data-driven computational tools for decision-making**
Theory: optimization, machine learning, control theory.
Methodology: machine learning for optimization, real-time optimization, optimization-based control, learning for control, statistical learning theory, computer-assisted optimization analysis.
Applications: autonomous systems, signal processing, robotics, power systems, data science, operations research, finance.

Publications

Journal Articles

- J1 R. Sambharya and B. Stellato, "[Learning Algorithm Hyperparameters for Fast Parametric Convex Optimization](#)," *SIAM Journal on Mathematics of Data Science*, vol. 8, no. 3, pp. 649–676, 2026.
- J2 R. Sambharya and B. Stellato, "[Data-Driven Performance Guarantees for Classical and Learned Optimizers](#)," *Journal of Machine Learning Research*, vol. 26, no. 171, pp. 1–49, 2025.
- J3 R. Sambharya, G. Hall, B. Amos, and B. Stellato, "[Learning to Warm-Start Fixed-Point Optimization Algorithms](#)," *Journal of Machine Learning Research*, vol. 25, no. 166, pp. 1–46, 2024.

Conference Proceedings

- C1 R. Sambharya, G. Hall, B. Amos, and B. Stellato, "[End-to-End Learning to Warm-Start for Real-Time Quadratic Optimization](#)," in *Proceedings of The 5th Annual Learning for Dynamics and Control Conference*, ser. Proceedings of Machine Learning Research, vol. 211, PMLR, 2023, pp. 220–234.

Preprints

- P1 R. Sambharya, N. Matni, and G. Pappas, "[Verification of Sequential Convex Programming for Parametric Non-convex Optimization](#)," 2025.

- P2** R. Sambharya, J. Bok, N. Matni, and G. Pappas, “[Learning Acceleration Algorithms for Fast Parametric Convex Optimization with Certified Robustness](#),” 2025.
- P3** A. Askari, G. Negiar, R. Sambharya, and L. E. Ghaoui, “[Lifted Neural Networks](#),” 2018.

Thesis

- T1** R. Sambharya, “Learning to accelerate optimization algorithms with guarantees,” Ph.D. dissertation, Princeton University, 2024.

Working Papers

- W1** R. Sambharya, S. Anand, and G. Pappas, *Verifying Performance, Stability, and Recursive Feasibility of Non-linear Model Predictive Control*.
- W2** S. Anand, S. Noorani, R. Sambharya, and G. Pappas, *Conformal Performance Guarantees for Learning to Optimize Mixed-Integer Programs*.

Honors and Awards

- Princeton Excellence in Teaching Award:** Top award winner in engineering 2021
- Princeton McGraw Teaching Fellow:** Led orientation for new teaching assistants 2022-2023
- Princeton SEAS Travel Grant Award:** (INFORMS) 2023

Talks

Verification of Sequential Convex Programming for Parametric Non-Convex Optimization

- INFORMS Optimization Society Atlanta, Georgia, March 2026

Learning Algorithm Hyperparameters for Fast Parametric Convex Optimization with Certified Robustness

- SIAM Conference on Optimization Edinburgh, Scotland, June 2026
- Industrial and Systems Engineering Department Seminar Texas A&M University, May 2026
- Industrial and Systems Engineering Department Seminar North Carolina State University, January 2026
- INFORMS Annual Meeting Atlanta, Georgia, October 2025
- International Conference on Continuous Optimization Univ. of Southern California, July 2025

Data-Driven Performance Guarantees for Classical and Learned Optimizers

- Neurips San Diego, California, December 2025
- International Symposium on Mathematical Programming Montreal, Canada, July 2024
- Optimization Learning and Control Workshop (Poster) Princeton University, June 2024
- INFORMS Optimization Society Rice University, March 2024
- Conference on Information Sciences and Systems Princeton University, March 2024

Learning to Accelerate Optimizers with Guarantees

- Thesis defense Princeton University, September 2024
- George Pappas's group** University of Pennsylvania, May 2024
- REALM lab** MIT, March 2024
- Computational Robotics Group** Harvard University, March 2024

Learning to Warm-Start Fixed-Point Optimization Algorithms

- Yale Robotics Seminar Yale University, December 2023
- INFORMS Annual Meeting Phoenix, AZ, October 2023
- Modeling and Optimization: Theory and Applications Lehigh University, August 2023

End-to-End Learning to Warm-Start for Real-Time Quadratic Optimization

- Learning for Dynamics and Control (Poster) University of Pennsylvania, June 2023

- NYC Operations day (Poster) Columbia University, May 2023
- INFORMS Annual Meeting Indianapolis, IN, October 2022

Accelerating Non-Convex Optimization via Learned Sequential Convexifications

- International Conference on Continuous Optimization Lehigh University, July 2022

Learning for Real-Time Semidefinite Optimization

- INFORMS Annual Meeting Anaheim, CA (hybrid), October 2021

Teaching

Texas A&M University (as Instructor)

- DAEN429: Predictive Analytics II Fall 2026

Princeton University (as Teaching Assistant)

- ORF499: Senior Thesis Spring 2024
- ORF498: Senior Thesis Fall 2023
- ORF363: Computing and Optimization for the Physical and Social Sciences Spring 2023
- ORF387: Networks Fall 2022
- ORF522: Linear and Nonlinear Optimization (Graduate-level) Fall 2021
- ORF307: Optimization Spring 2021 (Head TA), 2022
- ORF455: Energy and Commodities Markets Fall 2020

Mentoring

- **Research supervision:** Mentored graduate students: Tesshu Fujinami, Sima Noorani (Penn) 2025-
- **Research group leader:** Led weekly research group of 11 senior thesis students (Princeton) 2023-2024
- **Teaching assistant orientation leader:** Led yearly new TA orientation (Princeton) 2022-2023

Software

- Verification of Sequential Convex Programming for Parametric Non-convex Optimization
https://github.com/rajivsambharya/verify_scp
- Learning Acceleration Algorithms for Fast Parametric Convex Optimization with Certified Robustness
https://github.com/rajivsambharya/learn_algo_steps_robust
- Learning Algorithm Hyperparameters for Fast Parametric Convex Optimization
https://github.com/stellatogrp/learning_algorithm_hyperparameters
- Data-Driven Performance Guarantees for Classical and Learned Optimizers
https://github.com/stellatogrp/data_driven_optimizer_guarantees
- Learning to Warm-Start Fixed-Point Optimization Algorithms
<https://github.com/stellatogrp/l2ws>
- End-to-End Learning to Warm-Start for Real-Time Quadratic Optimization
https://github.com/stellatogrp/l2ws_qp

Industry Experience

- Machine Learning Engineer at Linc Global Sunnyvale, CA, July 2018 - July 2019
- Software Engineering Intern at Amazon Seattle, WA, June 2016 - August 2016

Service

- INFORMS Optimization Society Session Organizer: Data-Driven Algorithm Design and Analysis for Parametric Optimization 2026
- Group Meeting Organizer for the weekly Pappas Group Meeting 2025-
- Program Committee Member for the **ScaleOpt workshop** at Neurips 2025
- INFORMS Optimization Society Session Organizer: Machine Learning for Optimization 2024
- Princeton Optimization Seminar Organizer 2022-2023

Peer Review

- IEEE Transactions on Automatic Control (x1)
- Neurips (x2)
- IEEE Conference on Decision and Control (x1)
- Conference on Learning Theory (x2)
- Learning for Dynamics and Control (x5)
- Integer Programming and Combinatorial Optimization (x1)

Technical Skills

- **Programming languages:** Python, Matlab, Julia, R, C, Java, SQL, HTML
- **Tools:** Git, L^AT_EX, Slurm, GPU, JAX, PyTorch, Tensorflow